Labour-Market Adjustment and Fiscal Incidence of Trade Exposure in the UK*

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Abstract

For the same aggregate earnings loss, the UK tax–benefit system cushions an intensive-margin adjustment far more than an extensive-margin one. Using the 2024–25 Family Resources Survey and PolicyEngine UK, I hold a common sector-specific earnings loss of roughly £886 million a year, calibrated from the 2025 US tariff schedule, fixed while varying its incidence across incumbent wage reductions, displacement, and mixed transition paths. Proportional wage cuts are cushioned at 43.9 per cent, displacement at 36.8 ± 4.9 per cent: a gap of 7.3 percentage points (household-bootstrap 95 per cent interval 5.8–10.7 points), because in-work losses are relieved automatically at marginal tax rates while job loss relies on gated, means-tested, imperfectly claimed Universal Credit. An illustrative interior transition path yields 41.8 per cent cushioning and an Exchequer cost of £464 million. Sensitivity analysis quantifies the pension-saving component of the gap and varies adjustment shares, spell-duration equivalents, worker selection, and benefit take-up.

Keywords: tariffs, trade shocks, microsimulation, poverty, inequality, public finances

JEL codes: F13, F16, H23, H24, I38, J65

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1 Introduction

Who bears a trade shock when the same aggregate labour-income loss can be distributed through different labour-market margins? This paper answers that question for the United Kingdom. I hold a trade-exposure-calibrated loss in employee earnings—roughly £886 million a year, built from the 2025 US tariff schedule—fixed, and compare its fiscal and distributional consequences when it arrives as incumbent wage reductions, as displacement, or as mixtures of the two with re-employment penalties. The central finding is a large statutory asymmetry. Defining the cushioning rate as $1 - \Delta Y/\Delta E$, where ΔE is the gross employee-earnings loss and ΔY the household disposable-income loss, proportional wage cuts are cushioned at 43.9 per cent while displacement is cushioned at 36.8±4.9 per cent—a gap of 7.3 percentage points, with a household-bootstrap 95 per cent interval of 5.8 to 10.7 points that excludes zero. The mechanism is institutional: an in-work earnings reduction is offset automatically at marginal income-tax and National Insurance rates, whereas whole-job loss is relieved nearer average rates and depends on Universal Credit, which is gated by means tests, capital rules, and claiming behaviour. The gap is not an artefact of the HBAI treatment of pension saving: recomputed under a pension-gross income concept, both margins' cushioning falls by roughly four percentage points but the wage-minus-displacement contrast is essentially unchanged (Section 4.2).

The comparison matters because the UK tax–benefit system is nonlinear. For an equal aggregate loss, a broad reduction in earnings leaves most households in work and inside the tax schedule, whereas a concentrated loss moves households across benefit-eligibility thresholds. Fiscal incidence therefore depends on who loses earnings, whether the loss is temporary, and whether employment resumes—and a single wage-cut or displacement scenario should not be read as a forecast of realised trade adjustment. The two poles also price a structural disagreement: flexible-wage models turn a tariff primitive into a wage shock (Ignatenko et al., 2025), while downward-nominal-rigidity models turn the same primitive into a job shock (Rodríguez-Clare et al., 2025). Rather than endogenising the margin, I hold the aggregate loss fixed and report what each pole implies for households and the Exchequer.

The epistemic status of every input is stated once, here, and in the design table of Section 3.1, rather than repeated throughout. The estimand is conditional: the statutory tax–benefit and distributional response to a specified earnings-loss path, not a causal estimate of the 2025 tariff episode. The 2025 US tariff schedule supplies sector weights and shock magnitudes through a transparent accounting bridge (tariff rate × US-export intensity × a demand calibration × a wage-bill-incidence assumption); the bridge deliberately omits prices, quantities, imported inputs, productivity, profits, and firm entry. Reported ± values are assignment dispersion from the Monte Carlo allocation of displacement, not standard errors; survey-sampling uncertainty is scored separately by a household bootstrap. Mixed-path shares are disciplined by external evidence but not estimated.

One negative result should be on the table from the outset. A monthly HMRC product–destination panel confirms a post-policy decline in US-bound exports, but at sector rank the calibration fails its own falsification test: the rank correlation between predicted and realised sector export falls among the ten largest exposed divisions is zero (Section 4.3). This failure bounds the paper's claims—the sector calibration must not be read as measured tariff incidence—but it does not touch the cushioning comparison, which conditions on a declared loss rather than on the bridge. The leave-one-division-out audit across all 23 exposed divisions shows the wage-minus-displacement contrast surviving every rerun (range 5.3–9.6 points): the calibration determines *who* is shocked, not the statutory asymmetry between intensive and extensive relief.

The second stage uses the 2024–25 Family Resources Survey and PolicyEngine UK to compute taxes, National Insurance, benefits, household disposable income, poverty, inequality, and Exchequer effects. I report a common aggregate loss under alternative adjustment paths, decompose the result by tax and benefit channel, and show sensitivity to spell duration, worker selection, benefit take-up, and re-employment. Between the polar bounds, an illustrative interior transition path—70 per cent of the loss borne by workers who re-enter services jobs at an FRS-calibrated earnings penalty after a three-month lag, the rest by survivor wage cuts—yields 41.8 per cent cushioning and an Exchequer cost of £464 million. The response surface around the bounds, not any single interior point, is the paper's central object.

The paper makes three contributions. First, it provides a reproducible UK microsimulation framework for mapping trade-exposure-calibrated earnings losses into fiscal and distributional outcomes. Second, it shows why the incidence margin matters even when aggregate lost earnings are identical. Third, it quantifies which statutory channels drive the differences and which assumptions—timing, take-up, pensions, and worker selection—are most important for policy conclusions. The remainder of the paper describes the evidence and scenario design, presents the polar bounds and mixed paths, reports policy counterfactuals, and discusses limitations and reproducibility.

2 Literature

This paper sits at the intersection of four literatures: the incidence of trade shocks; the evidence on how workers adjust to them; the institutional analyses of the 2025 US tariffs on the UK; and the microsimulation literature on tax–benefit cushioning of earnings shocks. Existing macro–micro studies impose labour-market shocks on tax–benefit models, while tariff studies map price and income exposure into heterogeneous household welfare—the World Bank's Household Impacts of Tariffs framework combines household consumption and income portfolios (Artuc et al., 2021), and the OECD links its METRO trade model to household-budget surveys (Luu et al., 2020). The narrower contribution here is a UK application that maps the 2025 partner-country export-demand exposure into alternative labour-income stress scenarios and then through the statutory tax–benefit system; relative to those economy-wide tools it supplies only the terminal tax–benefit module, conditioning on an imposed labour-income distribution rather than deriving it from production (see Section 3.1).

2.1 Trade shocks and their incidence

Two recent papers anchor the structural end and frame the adjustment-margin question as its poles. Ignatenko et al. (2025) build a quantitative general-equilibrium model of the 2025 tariffs in which wages are flexible and labour supply elastic, so employment is an equilibrium outcome whose sign flips with revenue recycling; their replication data provide the UK-specific aggregate anchors I use. Atkin et al. (2025) instead apply a wedge framework to Chilean administrative data with factor supplies fixed: all adjustment runs through flexible wages and prices, and their "statutory versus economic incidence" framing is the one I adopt. Between the poles, Rodríguez-Clare et al. (2025) embed the 2025 trade war in a model with downward nominal wage rigidity, in which rigid wages turn the same tariff primitive into an employment shock that flexible-wage models turn into a wage shock—precisely the disagreement my adjustment-margin axis makes an explicit scenario dimension. Michelena et al. (2025) find 2025 employment and export losses concentrated in the targeted countries, and den Besten and Känzig (2025) show, in a narrative series over 1840–2024, that tariff increases are robustly contractionary—discipline for treating the shock as a negative export-demand shock.

The reduced-form lineage begins with Autor et al. (2013): import competition caused large, persistent employment and wage losses in exposed US commuting zones (effects persisting to 2019; Autor et al., 2021). Caliendo et al. (2019) provide the structural counterpart, Adão et al. (2023) the mapping from regional estimates to aggregates, Kim and Vogel (2021) the money-metric worker welfare translation with regressive incidence, and Galle et al. (2023) the coexistence of aggregate gains with group-specific losses. My contribution relative to this work is to replace stylised worker groups and transfers with real households and the actual UK tax–benefit system. The 2018–19 US episode supplies the sharpest evidence on who pays a tariff: pass-through into US import prices was essentially complete (Amiti et al., 2019; Fajgelbaum et al., 2020), retail prices adjusted more slowly (Cavallo et al., 2021, 2025), and US manufacturing employment fell on net (Flaaen and Pierce, 2019). The retaliation side is the closest analogue to my channel—a partner's tariff arriving as an export-demand shock (Waugh, 2019; Blanchard et al., 2019). The broader distributional literature establishes redistribution along both consumption and income margins (Fajgelbaum and Khandelwal, 2016, 2022; Borusyak and Jaravel, 2021). Two propagation channels my direct-channel design excludes have well-measured analogues: supply-chain transmission (Barrot and Sauvagnat, 2016; Acemoglu et al., 2016) and local-services spillovers (Moretti, 2010; Gathmann et al., 2020); Section 6 carries the caveat. Exchange rates absorb part of a tariff, imperfectly and state-dependently (Jeanne and Son, 2024; Corsetti et al., 2025; Kalemli-Özcan et al., 2026; Auray et al., 2020); I hold the exchange rate fixed. Early assessments of the 2025 wave again find the incidence landing on the tariff-imposing country's consumers (Fajgelbaum and Khandelwal, 2026)—complementary to my question, the incidence on the tariffed *partner's* workers.

2.2 Adjustment margins: how workers absorb trade shocks

The worker-level evidence says the margin is mostly earnings, not permanent unemployment. Autor et al. (2014) find exposed US workers lose earnings amid churn rather than indefinite joblessness; Traiberman (2019) shows occupation-specific human capital produces reallocation with persistent wage losses. Because my shock is an export-demand shock rather than import competition, the most direct precedents are Hummels et al. (2014) and Garin and Silvério (2024), who show exporters pass export-demand shocks into incumbent wages—evidence for including a wage-cut dimension. The rent-sharing literature disciplines its plausible scale: elasticities of wages to firm-specific rents of 0.05–0.15 (Card et al., 2018; Kline et al., 2019). Dauth et al. (2021) provide the cleanest evidence for the reallocation family, Dix-Carneiro (2014) the adjustment-cost horizons and age gradients, and Rud et al. (2022) decompose displacement costs into wage cuts on re-employment versus non-employment spells—directly calibrating the split between my families.

The UK's own history points to a third margin: benefit-financed economic inactivity. Beatty et al. (2007) document that coalfield job losses became male inactivity on incapacity benefits, and Beatty and Fothergill (2017) show that 1980s–90s industrial job destruction still inflates the deficit—the direct UK precedent for my fiscal-incidence question, retrospective and area-level where mine is prospective and household-level. The margin is live: disability claims still respond to local slack (Roberts and Taylor, 2022), hidden unemployment persists (Beatty and Fothergill, 2023), and working-age health-related benefit rolls reached 4.2 million by 2024 (IFS, 2024). For UK-specific elasticities, De Lyon and Pessoa (2021) show workers in Chinese-import-competing industries lost employment years and earnings (see also Dorn and Levell, 2024; Aragón et al., 2018). Closest to my object, Irastorza-Fadrique et al. (2025) study UK household responses to trade shocks—added-worker effects, retirement timing, self-employment—but model behaviour, not tax–benefit mechanics; I treat their margins as complementary.

2.3 The 2025 tariffs and the UK

The institutional literature on the 2025 shock is rich but stops above the household. The OBR puts the permanent UK GDP loss from a 20-point US tariff at one-third to three-quarters of a per cent (OBR, 2025); NIESR provides NiGEM scenarios (NIESR, 2024). The Bank of England estimates the effective US tariff on the UK at roughly 9 per cent and finds trade diversion has mildly *lowered* UK import prices (Bank of England, 2025; Dhingra, 2025)—which licenses isolating the export-demand/earnings channel while holding price and monetary channels fixed. The IMF and WTO frame the aggregates (IMF, 2025; WTO, 2025); UK firm surveys find mostly small expected effects, with the US taking about 22 per cent of UK exports (CEP, 2025). The geography is sharp: Coventry sends 22 per cent of its exports to the US (Centre for Cities, 2025); auto tariffs are estimated to cost £10,000 million of GDP over five years, 62 per cent borne by the West Midlands (City-REDI, 2025); SMMT data show US vehicle exports collapsing between April and May 2025 (SMMT, 2025). Some 80 per cent of UK steel exports go to Europe—the basis of my steel caveat—and the December 2025 pharmaceutical deal was paid for through VPAG rebate cuts and NHS pricing concessions, making pharma a fiscal/pricing shock rather than an employment shock (ABPI, 2025). The Resolution Foundation's household-facing analysis is descriptive rather than microsimulated (Resolution Foundation, 2025a), and the IFS has no dedicated distributional publication on the 2025 tariffs; that gap defines the contribution. For UK trade-shock precedents, Dhingra et al. (2017) map Brexit trade costs to incomes, Fetzer and Wang (2024) provide district-level incidence, Fetzer (2019) and Bloom et al. (2019) the political-economy and uncertainty channels, and Breinlich et al. (2022) and Levell et al. (2017) benchmark the consumption-price channel that this cash-income exercise holds fixed.

2.4 Tax–benefit cushioning of earnings shocks

The methodological ancestors are Dolls et al. (2012), who show EU tax–benefit systems absorb about 38 per cent of a proportional income shock and 47 per cent of an unemployment shock. Their *UK* cells are the right comparator: a UK income-shock stabilisation coefficient of 0.352 (income tax 0.267, social insurance 0.054, benefits 0.031) and a UK unemployment-shock coefficient of 0.415 (tax 0.191, SIC 0.061, out-of-work benefits 0.163). My 43.9 per cent wage-cut and 36.8 ± 4.9 per cent displacement rates are of a similar order; Section 4.2 focuses on the *composition* of the total for the exposed-worker population. The marginal-rate framing of the income tax as an automatic stabiliser originates with Auerbach and Feenberg (2000) and is formalised by McKay and Reis (2016). Sutherland and Figari (2013) establish EUROMOD's stress-test use; my direct methodological sibling is Brewer and Tasseva (2021), who pass the Covid-19 labour-market shock through UKMOD—the same architecture, applied here to a trade-policy shock without the emergency schemes and with the adjustment margin as the central axis. The macro–micro linkage literature supplies the shock-derivation lineage (Bourguignon et al., 2008; Peichl, 2016; Navicke et al., 2013). On the US side, exposed commuting zones saw rising disability, retirement and welfare payments offsetting only a small fraction of losses (Autor et al., 2013), and

Hyman et al. (2024) show wage insurance raises re-employment—a candidate counterfactual reform for this framework. The nearest tariff-microsimulation links are US consumption-side models (Tax Policy Center, 2025; Budget Lab, 2025; Acosta and Cox, 2024). Two institutional facts motivate the cushioning asymmetry examined here: UC's capital rules—thresholds frozen in cash terms since 2006—reduced or extinguished entitlement for around two million income-eligible families (Resolution Foundation, 2025b), and take-up of income-related benefits is well below 100 per cent (DWP, 2024), so statutory replacement rates can overstate what displaced workers receive. Take-up is examined as a headline sensitivity (Sections 3.6 and 4.2); the capital rules are a real-world gate the plain FRS data cannot exercise (Section 6).

3 Methodology

3.1 Design: primitives first

Throughout, "derived" means generated by the stated scenario mapping, not causally estimated. The central demand parameter is a transparent unit stress, $\varepsilon = 1.0$, rather than an estimate. The grid includes $\varepsilon = 0.4$ as an OBR-style lower anchor, 2.0 as the former April-outturn high case, and 3.0 as a severe sensitivity. The April movement is not used to identify the centre because it also reflects anticipation, front-running, prices, exchange rates, composition and other demand conditions. Likewise, $\pi = 1.0$ is a deliberately strong full wage-bill-incidence assumption. The parameter grid is scenario sensitivity, not a statistical confidence interval.

A reader is entitled to ask, before any number is reported, which parts of this design are read off data and which are assumed. The answer, stated in full:

Object	Status
Tariff schedule τ_j	**Data** (policy documents: the April 2025 schedule, the May 2025 EPD, the December 2025 pharmaceutical agreement).
US-export intensity x_j	**Data** (HMRC RTS/OTS customs exports over ONS ABS division turnover; Appendix A).
Export fall	**Data** in the measured family (realised HMRC outturn, May 2025–February 2026); **assumed** in the tariff families through declared low, central, high and severe demand scenarios.
Pass-through π	**Assumed** ($\pi = 1.0$), varied on a grid (Appendix A).
Adjustment margin	**Assumed**, and deliberately varied. This is the paper's free dimension.
Household incidence	**Computed** from statutory tax and benefit rules in PolicyEngine UK; no behavioural or reduced-form step.

The margin is assumed because the literature does not agree on it, and the disagreement is structural rather than empirical: Ignatenko et al. (2025) derive the employment response from elastic labour supply and flexible wages, while Rodríguez-Clare et al. (2025) derive it from downward nominal wage rigidity, so the same tariff primitive arrives as a wage shock in the first model and a job shock in the second. I do not endogenise the margin. I price the disagreement—holding the aggregate earnings loss fixed and reporting what each pole implies for households and the Exchequer. That is a weaker claim than a structural model makes, and a more auditable one.

The shock enters through primitives, not through an assumed employment effect. The pipeline is:

$$s_j \;=\; \tau_j \, x_j \, \varepsilon \, \pi, \tag{1}$$

where s_j is an imposed employee-wage-bill stress for SIC 2007 division j, τ_j is its tariff rate, x_j is US-export intensity, ε is an export-demand calibration and π is a *wage-bill-incidence assumption*. Calling π pass-through would conflate distinct mechanisms. A structural transmission would separately map the tariff into export prices and quantities, quantities into domestic production and value added, and production into factor demand, productivity, profits, wages, hours and employment. Equation (1) compresses those links into one accounting bridge. It neither estimates a production function nor identifies the causal response of productivity, output or labour demand.

x_j is US goods exports as a share of division turnover, built from HMRC trade data and ONS Annual Business Survey turnover denominators; Appendix A documents the frozen provenance and denominator choice. The central $\varepsilon = 1.0$ is a transparent unit stress, not an elasticity estimate; it amounts to a unit tariff-to-export-demand semi-elasticity, under which each percentage point of tariff removes one per cent of US-bound demand, and $\varepsilon = 0.4$ is the lower anchor, in the spirit of the trade-elasticity assumptions underlying the OBR's March 2025 tariff scenarios (OBR, 2025). The central $\pi = 1$ assigns the full calibrated exposure to the employee wage bill; it is intentionally severe and is not established by the economics literature. The sensitivity grid varies $\varepsilon\pi$, but this is parameter sensitivity, not an uncertainty interval around an identified coefficient. The implied gross earnings loss is therefore a conditional scenario input. OECD METRO and the UK Trade Modelling Review instead place tariff shocks inside multi-sector production systems in which prices, output, value added and factor demand adjust jointly (Luu et al., 2020; Department for International Trade, 2021); the World Bank HIT framework similarly labels its direct household calculations first-order and excludes productivity and substitution responses (Artuc et al., 2021). This paper begins only after those missing production-side mechanisms would have generated a labour-income distribution.

3.2 Public product–destination benchmark

To test whether post-policy US exports move differently from other UK export destinations, I build a monthly HMRC OTS panel from January 2018 to May 2026 for the United States, Canada, Japan and Australia (HM Revenue and Customs, 2026). Detailed HS8 rows are aggregated to HS4 and balanced across products, destinations and months, with absent flows set to zero. HMRC's separate two-digit aggregate rows are excluded. For product p and month t, the outcome is

$$g_{pt} = \log(1 + V_{p,\mathrm{US},t}) - \frac{1}{3} \sum_{d \in \{\mathrm{CA},\mathrm{JP},\mathrm{AU}\}} \log(1 + V_{pdt}).$$

I regress this destination gap on a post-May-2025 indicator, HS4 product fixed effects, month-of-year effects and a linear trend. April 2025 is omitted as an announcement and anticipation month. The primary WLS weights are products' mean 2022–24 US export values, capped at the 99th percentile; standard errors are clustered by HS4. Uncapped value weights, equal product weights, individual destinations, later sample starts and pseudo-policy dates in May 2019–23 are reported as specification checks.

This design controls for product shocks common across destinations but not US-specific demand, exchange rates, shipping, inventories, composition or other coincident policies. The high-tariff comparison groups HS chapters 72, 73 and 87 as steel and automotive products; that broad mapping is a falsification check, not a complete product-level statutory tariff schedule. The panel is therefore not used to estimate ε or π in Equation (1). It tests whether the imposed bridge is consistent with destination-differential outturn patterns.

3.3 The tariff schedule and the EPD counterfactual

Two tariff scenarios bracket the policy. *Full tariffs*: the April 2025 schedule—10 per cent baseline on UK goods, 25 per cent on motor vehicles (SIC 29) and steel (SIC 24). *EPD*: the deal-mitigated schedule—autos at the 10 per cent in-quota rate on up to 100,000 vehicles; steel with conditional, partial relief; pharmaceuticals (SIC 21) exempt under the December 2025 agreement. The automotive scenario applies the 10 per cent rate to the modelled 2024 exposure because the quota is approximately equal to that year's US-bound volume; it does not model quota allocation, above-quota growth or an endogenous volume response around the ceiling. The difference between the two scenarios, run through identical downstream machinery, prices the EPD for households and the Exchequer. Two caveats travel with the schedule. Steel: about 80 per cent of UK steel exports go to the EU, so the US tariff is a minority driver of the sector's exposure and the steel cell should be read alongside EU/UK safeguards. Pharmaceuticals: the exemption was purchased through VPAG rebate cuts and NHS pricing concessions, so pharma's incidence is fiscal and price-side, not an employment shock; I model its employment shock as zero under the EPD and discuss the fiscal channel separately (Section 5).

3.4 Adjustment-margin families

The derived s_j is realised on Family Resources Survey 2024–25 employees in exposed divisions. The submission design treats the mixed adjustment surface as the conceptual centre: statutory cushioning is evaluated for a common expected sector wage-bill loss while the share delivered through temporary non-employment, survivor earnings reductions, and re-employment is varied. The pure displacement and proportional wage-cut cases are transparent bounds on that surface. Inactivity, literal sectoral reallocation and the EPD schedule are complementary sensitivities, not alternative causal estimates. Wage cuts remove $\sum_j s_j B_j$ deterministically; displacement removes that amount in expectation under the risk model and is numerically integrated using the balanced design below.

Displacement (ADH short run). Within each exposed division every employee has baseline risk s_j, irrespective of the record's survey weight. The primary estimator generates repeated random-start systematic candidates, ordered across age and within-industry earnings ranks, and retains the candidate that best matches the expected industry wage-bill and weighted- head-count losses while secondarily balancing broad age and earnings margins. This is balanced numerical integration, not a survey estimator: conditioning on aggregate balance intentionally changes exact record-level marginal inclusion probabilities, and because the retained candidate minimises a score dominated by the total wage-bill target, per-division losses are matched softly rather than exactly and best-of-candidates selection carries no formal unbiasedness guarantee. The claim that the balanced design removes the declared amount in expectation is therefore established empirically, by its agreement with the unbiased comparator, not by construction. Selected workers move to unemployment with earnings, hours, pension contributions and statutory pay zeroed. An independent Bernoulli estimator, which preserves the declared first-order probabilities exactly but has much larger allocation dispersion, is reported as the unbiased assignment-rule comparator; at the unit scale the two agree on cushioning to half a percentage point (Section 4.2).

Wage cuts with reallocation (Dauth/Traiberman long run). No job loss; every employee in division j takes a proportional cut of s_j, so the weighted aggregate earnings removed equals $\sum_j s_j B_j$ by construction—exact earnings-equivalence with the displacement family's expected removal. Employee pension and salary-sacrifice contributions are scaled proportionally with the cut, mirroring their zeroing under displacement; this lowers taxable pay (and hence tax relief) but also reduces the household's pension outflow, so under the HBAI disposable-income concept part of the earnings loss is absorbed by reduced retirement saving rather than by current income.

Inactivity (Beatty–Fothergill UK history). The displacement draw, but displaced workers aged 50 and over exit to economic inactivity, the benefit-financed route of UK deindustrialisation, rather than unemployment. The route is benefit-financed in the model, not just relabelled: every inactive worker is flagged as having limited capability for work-related activity, so their household's Universal Credit includes the LCWRA health element (£217.26 a month in 2026—the reduced rate applying to new claims from April 2026 under the Universal Credit Act 2025; existing claimants retain the higher pre-reform rate, and all modelled inactivity claims are new claims). In policyengine-uk 2.89.2 the survey-imputed `uc_LCWRA_element` is stored, so the post-shock benefit-unit value is set to the maximum of the stored amount and one annual statutory element; an existing recipient is never paid a second element, unaffected units are unchanged, and the run hard-errors if the element fails to reach a newly flagged person's benefit unit. This mirrors the Beatty et al. (2007) route by which industrial job loss became incapacity-benefit inactivity. The strong assumption should be stated plainly: *all* displaced workers aged 50 and over are assumed to pass the Work Capability Assessment, so the family is an upper bound on the health-element route; no partial-qualification sensitivity (for example, 50 per cent of the older displaced qualifying) has been simulated, which is a limitation of the present scenario set. Work-search conditionality carries no separate fiscal machinery in the model, so the LCWRA element is the entire modelled wedge between inactivity and unemployment.

Reallocation into services (ADH/Autor et al., 2014). The same displacement draw on the same seed—so the two families are paired worker-for-worker—but the drawn workers are re-employed rather than left out of work. Their `sic_industry_division` is reassigned to one of four services destinations (SIC 47 retail, 86 human health, 49 land transport, 56 food and beverage service) in proportion to those divisions' FRS employee-weight shares (29.5, 45.8, 9.2 and 15.5 per cent), and their annual earnings are cut by a penalty calibrated on the FRS itself: weighted mean annual employee earnings of £48,272 over 2,047 hours in the exposed goods divisions against £34,594 over 1,701 hours in the four destinations, a **28.3 per cent annual-earnings penalty** that embeds the hours fall. Two variants are reported: an hours- and age-controlled **low-penalty** case of 14.0 per cent (the pure hourly-wage gap), and a **three-month lag** in which workers are out of work for a quarter of the year before the services job starts, scaling annual earnings and hours by a further $1 - 3/12$. ASHE 2024 full-time medians bracket the hours-controlled figure at 10–20 per cent; the FRS all-employee penalty is larger because it also captures the shift into part-time work. Because this family removes only the penalty on the movers rather than the full earnings-equivalent aggregate, its pound figures are comparable with the displacement family (identical worker sets, paired draws) but *not* with the wage-cut family; its cushioning *rate* is comparable with both. Appendix A reports the results and mechanics.

Common-loss transition path. The redesigned transition margin combines a transition-equivalent annual earnings/hours penalty, service-sector re-employment and survivor earnings reductions while preserving the common-loss normalization. Let s be the sector shock, λ the share of the declared loss assigned to transition workers, and $d = 1 - (1-r)(1-\ell/12)$ the annual earnings loss of a transition worker, where r is the re-employment penalty and ℓ the lag in months. Transition probability is $p = \lambda s/d$; survivors take a cut $c = (1-\lambda)s/(1-p)$. Hence $pd + (1-p)c = s$ in expectation. The illustrative interior calibration uses $\lambda = 0.70$, $r = 0.283$ and $\ell = 3$; the transition share is bounded by the displacement-cost decompositions of Rud et al. (2022) and Dix-Carneiro (2014), which attribute the majority of displacement costs to non-employment spells and re-employment penalties rather than survivor wages, but it is an assumption, not an estimate. It is one point on the response surface, which remains the paper's central object. In the annual simulation, the lag is represented by an earnings/hours factor: workers remain employed for PolicyEngine purposes, so no within-year unemployment or JSA/UC spell is generated. This is an accounting normalization for comparing paths, not an estimate of the probability of re-employment or the UK tariff response. The 14 per cent hours/age-controlled penalty and zero-, three- and six-month lags are sensitivity dimensions.

Mixed adjustment and factorial scenario testing. Let $\lambda \in [0,1]$ denote the share of the sector shock delivered through temporary non-employment rather than an incumbent earnings reduction. A worker with sector shock s is displaced with probability $p = \lambda s$. Conditional on surviving, the proportional wage cut is

$$c = \frac{s-p}{1-p},$$

so expected earnings loss is exactly $p + (1-p)c = s$ for every worker. The endpoints reproduce pure wage cuts ($\lambda = 0$) and pure temporary displacement ($\lambda = 1$); because the mixed family draws displacement independently, the $\lambda = 1$ endpoint reproduces the Bernoulli comparator rather than the balanced primary estimator. A separate reallocation family moves displaced workers into observed service destinations, with either an immediate earnings penalty or a three-month non-employment lag. It is paired worker-for-worker with displacement but does not preserve the common aggregate loss unless its penalty is explicitly re-scaled, so its pound totals are reported as a transition sensitivity rather than combined with the mixed grid. The exploratory grid crosses $\lambda \in \{0, .25, .5, .75, 1\}$ with export-demand calibrations $\varepsilon \in \{0.4, 1, 2, 3\}$ on five common assignments per cell. This follows the companion UK AI study's factorial principle: vary shock magnitude separately from adjustment composition. It is scenario testing, not a probability distribution over future outcomes.

The wage-cut margin as an upper bound, and a literature-disciplined mixed sensitivity. The pure wage-cut family should be read against the rent-sharing evidence, because it embeds a survivor-wage response far above anything estimated. Delivering the entire sector wage-bill loss through proportional cuts to incumbent workers' wages imposes a unit wage response to the sector exposure index. Garin and Silvério (2024) estimate an incumbent-wage elasticity of roughly 0.13–0.14 to firm-specific output shocks in Portuguese matched data; the rent-sharing survey of Card et al. (2018) places typical elasticities at 0.05–0.15; and Hummels et al. (2014) find small within-job wage responses in Danish data alongside losses through displacement. The wage-cut family ($\lambda = 0$) is therefore an *upper-bound stress scenario for survivor-wage incidence*, retained because it prices the flexible-wage pole of the structural disagreement.

The scenario surface also contains a 15/85 point ($\lambda = 0.85$): 15 per cent of the imposed wage-bill loss delivered through survivor wage cuts, disciplined by the upper end of the cited wage-response estimates. A wage elasticity to firm output or rents is not algebraically the share of an aggregate wage-bill loss borne by survivor wages—identifying that share requires linked employer–worker data—so this is one surface point among others, subordinate to the transition path above as the paper's single illustrative interior calibration.

Two scope limits on the family set should be stated here rather than in a footnote. The wage-cut family models the *earnings consequence* of reallocation without moving anyone between sectors: workers keep their observed industry and hours, and the proportional cut stands in for the wage penalty that a completed reallocation would carry; the reallocation family does move them, but who reallocates is assumed (it is the displacement draw) rather than estimated. And no household-side behavioural margin is active—added-worker responses, earlier or later retirement, and shifts into self-employment (Irastorza-Fadrique et al., 2025) are all suppressed, as are any labour-supply responses to the tax and benefit changes themselves. All four families are therefore first-round, rules-based incidence.

A third caveat concerns selection within divisions. Export-exposed workers are unlikely to be representative of their division: exporters pay a wage premium and pass export-demand shocks into incumbent wages (Hummels et al., 2014; Garin and Silvério, 2024). Cushioning is nonlinear in earnings, so uniform risk may misstate its tax/benefit composition. The longitudinal-LFS cell, earnings-band and QRF sensitivities vary observed risk shape while recalibrating each division to its wage-bill target. They quantify selection- model sensitivity but cannot identify exporter status or tariff-specific worker risk.

The FRS support for the transition is thin. Of 34,966 person records, 12,800 have employee earnings and 1,018 are in mapped exposed divisions, but the reference probabilities imply only 9.7 selected records. The largest exposed record has a person weight above 19 thousand. Balanced integration constrains the realised aggregate and observed composition; it does not create survey information or establish population representativeness. I therefore report the affected record count, Kish-style effective number of gross-loss contributions, largest-record loss share, and balanced-versus-Bernoulli mean and SD comparisons. Subgroup rankings remain exploratory.

Duration-equivalent annual stresses. The annual PolicyEngine model cannot represent a worker employed for part of a year and unemployed for the remainder with internally consistent monthly UC assessment, work status and earnings. I therefore do not simulate three- or six-month unemployment spells. Instead I scale s_j by $d/12$, for $d \in \{3, 6, 12\}$, and assign a correspondingly smaller number of coherent full-year displaced states. These rows preserve the expected worker-month earnings loss associated with each duration while changing the extensive-margin stock. They are labelled *duration-equivalent annual stresses*, not spell estimates. Wage-cut rows use the same scale, preserving the common- loss comparison at every duration and at both the OBR-style 0.4 and unit anchors. Appendix K bounds the bias from this annualisation with a parameter-based comparison of the duration-equivalent stress against a correct monthly UC assessment for a representative displaced worker: UC entitlement per worker-month is identical by construction, the primary 12-month contrast is unaffected, and the residual differences at 3- and 6-month scales run through partial-year income tax and payment timing.

The margin is the paper's central axis because the tax–benefit system treats the two poles asymmetrically: an in-work earnings reduction is offset automatically at marginal tax and taper rates, while a job loss moves the worker to the gated, means-tested extensive margin of the benefit system. How much each pole is cushioned, and—the question that turns out to discriminate between them—how much that cushioning depends on anyone doing anything, are empirical questions the microdata answer (Section 4.2).

3.5 Longitudinal LFS imputation benchmark

The uniform within-division assignment in the central displacement family is transparent but suppresses observable worker heterogeneity. I therefore use UK Data Service Study 9490, the Labour Force Survey Five-Quarter Longitudinal Dataset covering July 2024–September 2025, as a donor benchmark. The donor population comprises respondents employed at wave 1. Job exit is observed at wave 5; log wage change is defined only for respondents employed with positive main-job pay at both endpoints. Negative UKDS sentinel values are missing, not zero. Public 2024 Nomis BRES employee totals rescale the manufacturing SIC composition before direct transition targets are computed. (Office for National Statistics, 2025)

The primary imputation maps SIC-division–sex–age cells to employed FRS receivers, shrinking sparse cells towards sector and manufacturing means. After matching, predicted exit risks are rescaled so their FRS-weighted mean equals the directly measured, BRES-composition-adjusted LFS manufacturing exit rate. Mean predicted log wage changes are centred on the corresponding LFS target. An income-tercile estimator supplies a transparent sensitivity specification.

A regularised quantile/random-forest benchmark follows the calibrated imputation architecture in PolicyEngine's `young-worker-nics` project. It estimates model shape on all wave-1 employed adult donors using age, sex and annualised employment income. The binary outcome uses the random-forest event probability; the continuous outcome uses the QRF conditional mean. Forests use 500 trees and a minimum leaf size of 20; wage outcomes are winsorised at the donor 1st and 99th percentiles. Both receiver predictions are then calibrated to exactly the same manufacturing LFS targets as the cell estimator. This is a predictive baseline-transition benchmark, not an estimate of tariff-induced exits or wage changes. It is therefore reported as a robustness diagnostic and does not replace the declared central assignment rule.

In a downstream allocation sensitivity, each receiver score r_i is converted to a within-division displacement probability $p_i = \min(1, k_j r_i)$, where k_j is solved numerically so

$$\frac{\sum_{i\in j} p_i E_i w_i}{\sum_{i\in j} E_i w_i} = s_j.$$

Thus cells, earnings bands and QRF change who is exposed while preserving each division's expected employee-wage-bill loss. Missing scores use the division mean. Common assignments then isolate sensitivity to incidence shape; they do not give the baseline transition model a causal tariff interpretation.

3.6 Universal Credit take-up after the shock

The take-up experiment is applied symmetrically across margins. After each shock, changed benefit units whose all-claim UC award moves from zero at baseline to positive post-shock receive a claiming draw at the scenario take-up rate. This includes wage-cut households that cross into entitlement. Existing claimants, existing eligible non-claimants, unchanged units and post-shock-ineligible units retain their baseline flag, avoiding an unintended population-wide take-up reform. The grid therefore measures sensitivity to a common new-entitlement convention; it does not estimate actual claiming behaviour.

One modelling choice deserves its own subsection, because an earlier version of this paper got it wrong; the correction turns out to leave the unit-stress headline unchanged (Section 4.2), but that is a measured result, not a licence for the incorrect convention. PolicyEngine UK carries benefit take-up as a benefit-unit-level *stored input*, `would_claim_uc`, drawn once at dataset build time. Two properties of that draw matter. First, its target is a flat population-wide parameter—0.55 as a share of *all* benefit units, with reported UC recipients anchored to TRUE and the remainder filled to hit the target. It is not a take-up rate among the entitled, and it is therefore not comparable to the roughly 80 per cent take-up-among-entitled that the Resolution Foundation and the IFS assume; in the dataset used here the realised weighted flag rate is 0.547 over all benefit units and 0.541 among UC-eligible ones. Second, the flag is drawn on *pre-shock* circumstances. Nothing in a naive shock pipeline re-draws it, so a worker displaced by the shock carries into unemployment the claiming decision attributed to them while they were employed and, in the great majority of cases, not entitled to anything—measured take-up among the displaced post-shock was 46.9 per cent, below even the population flag rate. The baseline flag is thus not informative about the post-shock claiming behaviour of a newly entitled family: it encodes a decision taken in a state of the world the shock has destroyed.

The pipeline therefore re-draws `would_claim_uc` after each shock only for benefit units whose circumstances changed and whose all-claim UC award moves from zero at baseline to positive post-shock, at a scenario parameter `uc_takeup` with default 0.80. The re-draw uses an independent random stream, leaving worker assignment unchanged and preserving paired comparisons. This rule applies to displacement, inactivity, reallocation and wage cuts alike; existing eligible, post-shock-ineligible and unchanged units retain their baseline flag. Because the entire displacement-side cushion runs through UC, take-up is treated as a headline sensitivity rather than an exploratory one: Section 4.2 reports the primary displacement margin at new-entitlement take-up rates of 0.55, 0.80 and 1.00 over the full comparator draw count. The 0.80 default is a strong assumption for a population with no UC claiming history and unobserved capital (redundancy payments would gate many claims under the frozen capital thresholds); the grid brackets it rather than estimating it.

3.7 Microsimulation and outcomes

The Monte Carlo draws describe sensitivity to the artificial assignment of displacement across weighted FRS records. Their standard deviations are assignment dispersion, not sampling standard errors, confidence intervals, or uncertainty about the tariff response, survey design or model parameters. Numerical Monte Carlo error of a reported mean is the assignment SD divided by the square root of the number of draws. Parameter scenarios vary the demand calibration, wage-bill incidence, take-up and adjustment composition, but are not assigned probabilities and do not form confidence intervals. Survey sampling variance is approximated for the primary contrast by a household bootstrap of the FRS sample (Section 4.2); the research file carries no PSU or stratum identifiers, so the bootstrap treats households as independent and cannot reflect geographic clustering. Uncertainty in the trade-to-earnings bridge is not estimated. Reported comparisons other than the bootstrapped primary contrast are descriptive scenario contrasts. Distributional cash-income changes use annual household disposable-income change divided by household size before person-weighted aggregation; aggregate disposable-income totals remain household-weighted totals.

The realisation step—an imposed status and earnings transition on survey households, then run through the full benefit rules—follows Brewer and Tasseva (2021), who apply it to the

Covid-19 labour-market shock in UKMOD, and the EUROMOD nowcasting tradition of imposed labour-market transitions (Navicke et al., 2013). PolicyEngine UK computes baseline and shocked household disposable income (HBAI cash concept throughout), relative before-housing-costs poverty measured against a poverty line frozen at 60 per cent of the baseline equivalised median (so the reported change reflects households crossing a fixed threshold rather than mechanical movement of the line itself), absolute before-housing-costs poverty against a fixed baseline line, after-housing-costs poverty, the Gini coefficient, distributional breakdowns and the Exchequer effect. The *Exchequer effect* is defined as the change in the aggregate simulated government balance—all modelled tax and contribution revenue (including employer National Insurance contributions) minus all modelled benefit spending—between the baseline and shocked simulations. It is therefore broader than the employee-earnings loss ΔE that enters the cushioning rate; Appendix A reconciles the two. On timing: trade exposure is built from 2024 HMRC customs data, the household input is FRS 2024–25, and the simulation year is 2026, with PolicyEngine UK's standard uprating applied to survey monetary variables between the survey year and the simulation year. Primary submission scenarios use 50 balanced assignments; legacy direct and measured scenarios use 100 assignments, and explicitly exploratory diagnostics use five. Following Ignatenko et al. (2025), revenue recycling can flip aggregate employment responses; here fiscal policy is held fixed. Only direct taxes and benefits respond. Indirect taxes are not shocked, so reduced consumption-tax receipts are absent from the government balance.

4 Results

All results use FRS 2024–25 and period 2026. The bounds table uses 50 balanced assignments; legacy secondary scenarios use 100. Values after $\pm$ are assignment standard deviations, not standard errors or confidence intervals. Numerical Monte Carlo SE is stored separately in each JSON artifact. Five-assignment diagnostics are exploratory.

4.1 Derived sector shocks

The deterministic OBR-style lower-anchor stress is £354 million annually. The unit-stress full-schedule amount is £886 million and the EPD unit stress is £693 million. These are scenario inputs generated by Equation (1), not estimated causal effects. The unit-stress sector rates are exactly half those of the former $\varepsilon = 2$ high case and 2.5 times the OBR-style rates.

4.2 Cushioning

Table 1 reports the two polar bounds of the incidence surface: it crosses the external OBR-style and deliberately stronger unit anchors with one common-loss contrast and three duration-equivalent scales. At the 12-month unit stress, wage-cut cushioning exceeds displacement cushioning by 7.3 percentage points (assignment SD 4.6), while its Exchequer cost differs by £83 million. These are conditional statutory comparisons, not confidence intervals for tariff effects. The mixed grid below is the conceptual centre of the design; the bounds make its composition effect transparent.

Assignment-design and record-support audit. At the unit 12-month stress, balanced and Bernoulli integration agree closely: cushioning is 36.6 versus 35.9 per cent and Exchequer cost is £401 versus £398 million. Balancing reduces gross-loss SD to 0.02 of the Bernoulli value and

Table 1: Primary common-loss contrast by anchor and duration-equivalent annual stress, 50 assignments

Anchor	Months	Margin	Gross loss (£m)	Exchequer (£m)	Cushioning (%)	Poverty (pp)
OBR	6	Displacement	177 ± 3	73 ± 13	32.7 ± 7.7	0.009
OBR	6	Wage cut	177	96	43.7	0.000
OBR	12	Displacement	354 ± 4	141 ± 21	32.0 ± 5.7	0.015
OBR	12	Wage cut	354	192	43.6	0.000
UNIT	3	Displacement	222 ± 4	91 ± 15	32.0 ± 6.2	0.008
UNIT	3	Wage cut	221	120	43.6	0.000
UNIT	6	Displacement	442 ± 6	190 ± 24	33.8 ± 5.4	0.016
UNIT	6	Wage cut	443	241	43.6	0.000
UNIT	12	Displacement	885 ± 6	378 ± 37	34.0 ± 4.7	0.036
UNIT	12	Wage cut	886	484	43.9	0.000

Notes: OBR uses export-demand elasticity 0.4; unit uses 1.0. Values after ± are assignment SDs. Displacement uses balanced numerical integration; wage cuts are deterministic apart from the common UC claiming redraw. Three- and six-month rows scale the annual probability of a coherent full-year displaced state; they match expected worker-month loss but do not simulate partial-year unemployment spells (Appendix K bounds the annualisation bias). The OBR three-month cell is omitted: its displacement draw rests on 1.2 effective record contributions and is reported in Appendix C as sensitivity evidence only. Poverty is the change in the person-weighted fixed-line national BHC rate; because the exposed population is small and largely above the poverty line, near-zero national changes carry no information about losses among affected households.

Exchequer SD to 0.19, but it cannot repair sparse survey support. The unit displacement loss has only 6.2 effective record contributions and its largest record supplies 28.7 per cent on average. At the OBR three-month scale these deteriorate to 1.2 and 91.9 per cent. The smallest displacement rows are consequently sensitivity evidence, not precise population estimates. Across leave-each-sector-out reruns for all 23 exposed divisions, the wage-minus-displacement cushioning difference remains positive, ranging from 5.3 to 9.6 percentage points. No single division creates or reverses the primary contrast.

Survey sampling uncertainty. Assignment SDs do not measure sampling variance, so the primary contrast is additionally scored with a household bootstrap of the FRS sample: 999 multinomial resamples of the 16,288 households, applied to per-household contributions from 25 common assignment draws of each unit 12-month margin, so that resampling variation is measured around the draw-averaged statistic. The wage-minus-displacement cushioning difference is 7.6 percentage points with a bootstrap standard error of 1.2 points and a 95 per cent percentile interval of 5.8 to 10.7 points: the interval excludes zero, so the sign of the primary contrast is not an artefact of which households the FRS happened to sample, although its magnitude is estimated imprecisely. Margin-level bootstrap SEs are 1.1 points (displacement cushioning), 0.9 points (wage-cut cushioning), and £45 million and £37 million on the respective Exchequer costs. The FRS research file carries no PSU or stratum identifiers, so households are the resampling unit; the interval approximates the design-based sampling variance without capturing geographic clustering, and it does not incorporate first-stage or parameter uncertainty, which remain scenario dimensions. One convention should be noted: a resample can zero a displacement draw's gross loss, leaving that draw's cushioning ratio undefined; replicates average over the defined draws, and near-zero-gross resamples with unstable ratios are retained. The interval is therefore a record-resampling sensitivity, not evidence that a conditional scenario contrast is statistically identified.

Pension-contribution channel. The HBAI disposable-income concept treats employee pension and salary-sacrifice contributions as an outflow, so a hypothesis worth testing is that part of the headline gap is reduced retirement saving rather than tax–benefit insurance. It is not. Recomputing both margins under a pension-gross income concept (disposable income plus employee pension and salary-sacrifice contributions, unit 12-month stress, Bernoulli comparator over 50 draws) lowers wage-cut cushioning from 43.9 to 39.6 per cent and displacement cushioning from 36.6 to 31.8 per cent—the pension outflow falls by close to the same share of the gross loss on both margins (4.4 and 4.9 per cent respectively; wage cuts scale contributions, displacement zeroes them)—so the wage-minus-displacement gap moves only from 7.3 to 7.8 percentage points. The headline contrast is a statutory tax–benefit asymmetry, not a pension-accounting artefact. What the pension channel does affect is the *level* of measured cushioning on both margins, and the implied loss of deferred retirement consumption is not captured by the cushioning rate.

Mixed and transition paths. The illustrative interior calibration is the transition-equivalent path: 70 per cent of each sector's declared loss is assigned to workers whose annual earnings and hours embody a three-month re-employment lag before entering a services job at the FRS 28.3 per cent earnings penalty, with survivor cuts rescaled so the expected sector loss remains common. It yields 41.8 ± 2.5 per cent cushioning and an Exchequer cost of £464 ± 136 million—between the bounds, as the composition logic implies. The surrounding response surface (Figure 1) includes a churn-dominated 85/15 displacement–survivor split (38.3 ± 4.2 per cent cushioning, £428 ± 221 million) and the lower 14 per cent penalty sensitivity reported in the appendix. The 70/30 share, lag and penalty are assumptions bounded by external evidence, not estimated tariff responses.

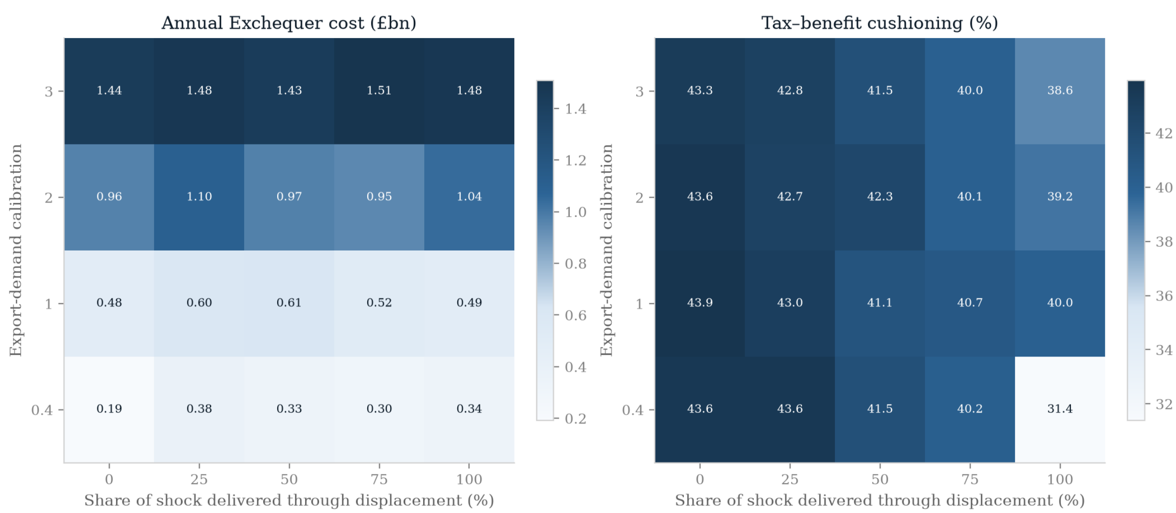

Figure 1: Factorial scenario surface: export-demand calibration by the share of expected sector loss delivered through displacement rather than survivor wage cuts. Cell values are means over five common assignments.

4.3 External magnitude and outturn checks

The frozen HMRC customs build totals £55.6 billion of 2024 US-bound goods exports before exclusions and maps £52.5 billion to producing divisions. This is reasonably close to the ONS balance-of-payments headline of £59.3 billion, given known conceptual and coverage differences (ONS, 2025). The reference full-schedule calibration predicts a 12.57 per cent aggregate fall across mapped divisions. The observed May 2025–February 2026 customs window falls 12.71 per cent against its two-year same-month baseline. This near equality is not a validation result: the reference calibration was not estimated from that window, and the outturn combines tariff effects with prices, exchange rates, anticipation, composition, rerouting and other demand shocks.

Sector agreement is much weaker. The stored validation artifact reports a rank correlation of 0.00 between the full-schedule predicted and realised falls among the ten largest exposed divisions, and a correlation of 0.58 between full-schedule and observed downside shock rates across all mapped divisions. Machinery and motor vehicles decline broadly in the expected direction, while aerospace, electrical equipment and basic metals do not. The aggregate match can therefore be coincidental and must not be used to validate Equation (1). As a separate policy benchmark, the OBR's March 2025 scenario uses a demand elasticity of 0.4, implying an 8 per cent export-demand fall after a 20 percentage-point tariff and a direct export-demand effect just under 0.2 per cent of GDP (OBR, 2025). The paper's $\varepsilon = 1$ reference is 2.5 times that elasticity and additionally assumes full wage-bill incidence; it is a deliberately strong stress, not a central forecast.

Product–destination panel. The monthly HMRC panel (Appendix H) supports a post-policy decline concentrated in large US export products—the capped value-weighted specification estimates a -33.9 per cent change in US-bound exports relative to the same products sent to Canada, Japan and Australia—but the tariff-intensity ordering fails: broad steel and auto chapters fall -22.5 per cent against -34.9 per cent for other products. As set out in the introduction, this failed falsification tests the bridge, not the estimand: the cushioning comparison conditions on a declared loss that is identical across margins by construction, and the leave-one-division-out audit (all 23 divisions, contrast range 5.3–9.6 points) together with the three LFS-shaped selection models confirm that sector composition does not create or reverse it. What the failure rules out is any reading of Equation (1) as an estimated first stage.

Longitudinal worker benchmark. The five-quarter LFS contains 177 wave-1 manufacturing employees after the required population filters, compared with 1102 employed manufacturing FRS receivers. Its BRES-composition-adjusted manufacturing targets are an 8.96 per cent wave-1-to-wave-5 job-exit rate and 5.87 per cent mean log wage growth among valid endpoint-pay observations. Table 2 compares receiver distributions. Calibration deliberately makes the aggregate exit means identical, so the comparison tests incidence shape rather than level. The QRF has a much wider risk distribution and agrees weakly with the primary cells: person-level correlations are only 0.06 for exit risk and 0.08 for predicted wage change. The longitudinal evidence therefore does not support replacing transparent cells with forest-generated tariff incidence. It instead quantifies model uncertainty that the central uniform assignment omits. Regularisation does not eliminate all forest tails: the maximum calibrated exit risk is 62.6 per cent and the predicted log-wage-change range is -45.7 to 72.0 per cent. These are benchmark predictions, not plausible confidence bounds.

Table 2: Calibrated FRS job-exit-risk distributions

Estimator	Weighted mean	P10	Median	P90
Calibrated SIC–sex–age cells	8.96	7.15	8.42	11.88
Calibrated income terciles	8.96	3.88	8.91	13.77
Regularised QRF benchmark	8.96	1.22	5.42	19.62

Notes: Entries are percentages among matched employed manufacturing FRS receivers. All means equal the same directly measured LFS target by construction. Quantiles are survey-weighted. QRF denotes the regularised quantile/random-forest benchmark described in Section 3.5; it predicts baseline transitions, not causal tariff effects.

Fiscal sensitivity to worker-risk shape. Table 3 uses each calibrated risk model to vary selection within SIC divisions while holding every division's expected weighted wage-bill loss at the original shock. The cell model nearly reproduces uniform assignment. Across all three LFS-shaped models, mean cushioning ranges from 36.0 to 37.4 per cent, Exchequer cost from £384 million to £425 million, and measured poverty change from 0.024 to 0.028 percentage points. This model spread is smaller than the assignment SD within any specification. Weighted displaced headcount changes because calibration preserves the wage bill, not a headcount quota. The QRF mean gross loss is £847 million rather than the £886 million expectation, but its £397 million assignment SD implies a numerical MC SE of about £40 million, so this difference is not evidence of a different aggregate target.

Table 3: Full-schedule fiscal sensitivity to within-SIC worker selection

Selection model	Workers (000s)	Gross loss (£m)	Exchequer (£m)	Cushioning (%)	Poverty (pp)
Uniform central assignment	17.2 ± 6.9	882 ± 418	387 ± 213	34.1 ± 5.2	0.032 ± 0.022
LFS calibrated cells	17.2 ± 6.9	881 ± 412	412 ± 223	36.9 ± 5.0	0.028 ± 0.019
LFS income terciles	15.2 ± 6.5	887 ± 467	425 ± 258	37.4 ± 6.0	0.025 ± 0.017
LFS regularised QRF	18.1 ± 5.9	847 ± 397	384 ± 212	36.0 ± 5.7	0.024 ± 0.016

Notes: Means ± assignment SD over 100 common assignments. Within each SIC division, individual probabilities are rescaled so the earnings-weighted mean equals the declared sector shock. Poverty is relative BHC against the frozen baseline line. LFS risks are predictive baseline-transition scores, not tariff-specific causal probabilities.

The realised displacement output is also consistent with the allocation design but too sparse for precise empirical incidence: the expected 9.7 selected FRS records per assignment produce 17.2 thousand represented displaced workers and £882 ± 418 million of gross loss. The gross-loss SD is nearly half its mean, and Exchequer-cost SD exceeds half its mean. The 100-assignment average is numerically more stable than an individual assignment, but it remains conditional on the observed 1,018 exposed records and the uniform within-division selection rule.

Take-up as a headline sensitivity. Because the displacement-side cushion runs through UC, take-up is scored at the full comparator draw count (50 Bernoulli assignments) rather than as an exploratory grid, with the new-entitlement claiming rate at 0.55, 0.80 and 1.00 and, additionally, the stale pre-shock claiming flag restored as a fourth convention. The measured answer is that none of these moves the unit-stress headline: displacement cushioning is 35.2 per cent under every convention, with differences below 0.1 percentage points—within the numerical precision of the aggregation. The reason is institutional and was not obvious ex ante: most displaced households are *already* entitled to UC at baseline through the in-work means test, so under the maintained rule they retain their stored claiming state, and the newly entitled population that the take-up parameter governs carries little survey weight. The residual convention that this grid cannot test is the retention of baseline claiming states for already-entitled non-claimants—about half of entitled benefit units in the input data do not claim—and the unmodelled UC capital test (redundancy payments against thresholds frozen since 2006), both of which would move measured cushioning down, not up (Section 6). Wage-cut results are take-up-insensitive because few wage-cut households change UC entitlement.

4.4 Mechanism

The regenerated decomposition and gate audit are five-assignment diagnostics, but they clarify why the two margins produce different fiscal outcomes. A wage cut keeps the worker inside the income-tax and National Insurance schedule, so relief is concentrated in lower tax and contribution payments. Displacement removes earnings altogether and shifts the response towards Universal Credit and other means-tested support. The latter channel is nonlinear: entitlement depends on household composition, existing income, rent, capital, and claiming behaviour rather than on the individual earnings loss alone. The plain FRS input contains no usable household-capital measure, so the model cannot establish the empirical importance of UC's capital test. New-style JSA is not activated by the imposed transition. The gate audit should therefore be read as a decomposition of the implemented rules, not as a complete account of the UK out-of-work system.

4.5 Distributional incidence

Under full displacement, relative BHC poverty rises by 0.028 ± 0.019 percentage points. Under wage cuts it is unchanged at displayed precision. The small national change does not imply that exposed households are protected: the affected population is limited, and many exposed workers begin above the poverty threshold. The decile profile shows a broad reduction in disposable income, with the largest losses concentrated towards the upper end of the distribution where exposed earnings are concentrated. This pattern is consistent with a progressive tax–benefit response that cushions lower-income households more strongly than the gross earnings measure. These figures are distributional diagnostics rather than population-wide poverty forecasts. Figure 2 shows the decile profile.

A constituency-level surface was produced in earlier drafts but is not reported: it rests on imputed industry assignments rather than observed local industry structure, and a map invites exactly the seat-ranking reading it cannot support. Appendix M documents the machinery for future use with observed local industry data.

4.6 Fiscal incidence

Annual Exchequer costs are reported in Table 1. The measure is the aggregate simulated government-balance change and is broader than the household cushioning identity. The cost combines lower income-tax and National Insurance receipts with additional benefit payments. Employer National Insurance contributions and the tax consequences of pension and salary-sacrifice changes respond to the shock but sit outside employee disposable income, which explains why the Exchequer cost does not equal the household-side cushioning amount. Displacement also produces more assignment dispersion because a small number of high-weight records can determine whether earnings are lost entirely. Appendix A provides the formal reconciliation and draw distributions. Figure 3 shows the assignment distribution.

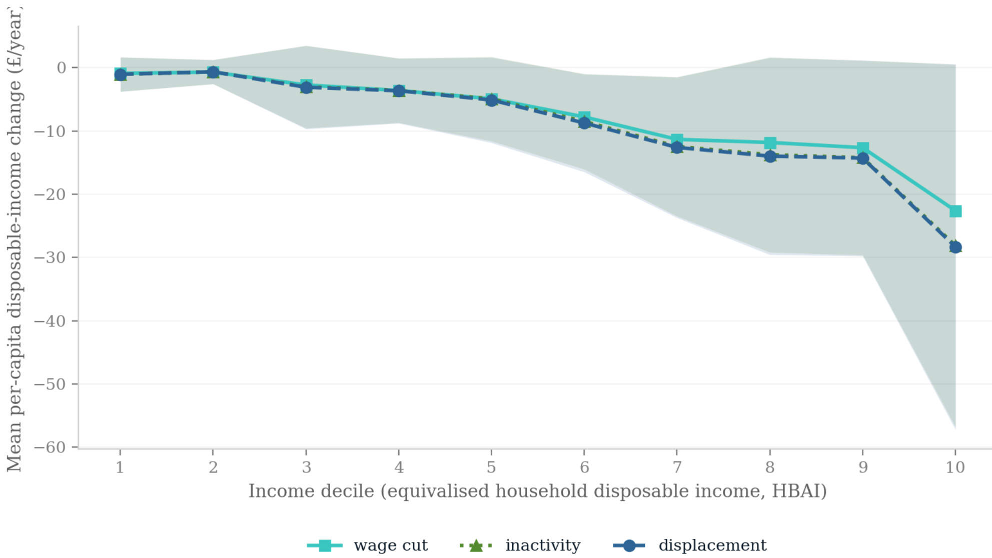

Figure 2: Per-capita disposable-income change by baseline income decile in the full-schedule scenarios. Shading denotes assignment dispersion.

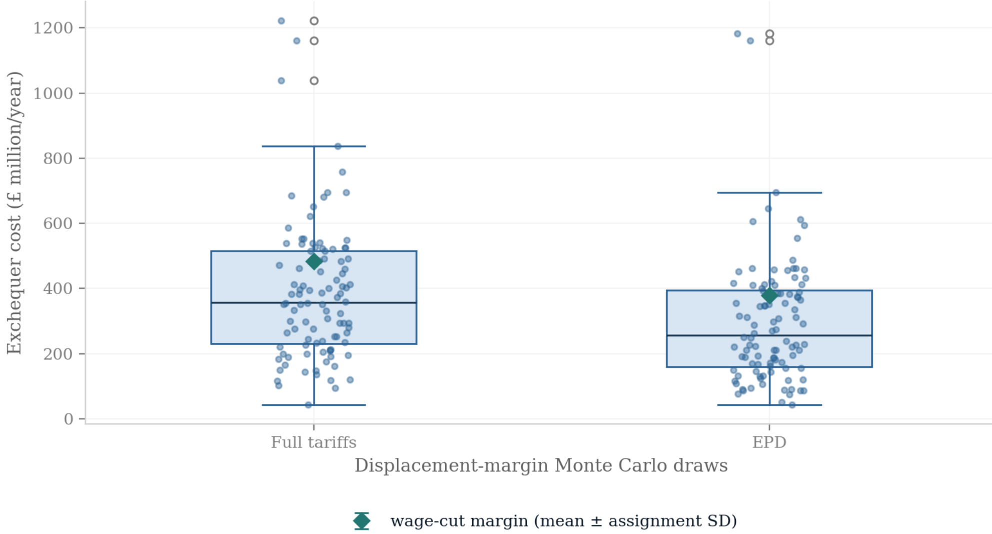

Figure 3: Annual Exchequer cost across regenerated assignments. The spread is assignment dispersion from the Monte Carlo allocation of displacement, not sampling uncertainty.

4.7 EPD counterfactual

On paired seeds, full minus EPD displacement differs by 3.7±2.8 thousand workers and £96±101 million of Exchequer cost. The contrast is driven primarily by the lower auto tariff and the exemption of pharmaceuticals in the EPD schedule. It should be interpreted as the fiscal value of changing the declared sector shock while holding the household model and assignment draws fixed. It does not include quota allocation, above-quota trade, price responses, NHS pricing concessions, or other general-equilibrium effects. Figure 4 summarises the paired comparison.

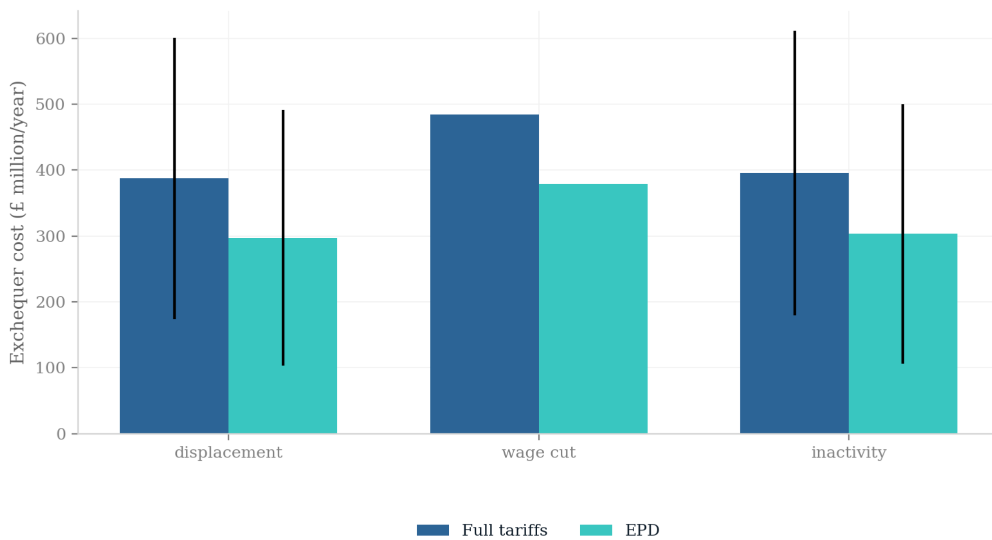

Figure 4: Full-schedule and EPD scenario comparison. Error bars are assignment standard deviations.

5 Policy

5.1 Economic Prosperity Deal

Within the maintained direct-channel scenarios, the paired full-minus-EPD displacement contrast is 3.7 ± 2.8 thousand workers, £204 ± 192 million of gross earnings and £96 ± 101 million of annual Exchequer cost. These SDs describe assignment dispersion and the contrasts are not causal estimates of the deal. Autos drive much of the difference; the scenario applies the in-quota rate to fixed exposure without modelling quota allocation or future above-quota trade.

5.2 Pharmaceutical channel

The EPD scenario sets the pharmaceutical employment shock to zero, but this does not make the exemption free. Its VPAG and NHS pricing concessions sit outside the labour-income microsimulation. A complete policy appraisal would cost those health-budget and price effects alongside the direct tax–benefit scenario.

5.3 Cushioning policy

The simulations support a limited conclusion: extensive-margin protection depends more on means-tested eligibility and claiming than the income-tax response to an intensive earnings loss. The full-draw take-up grid shows the claiming-rate assumption itself is not the binding parameter—most displaced households are already UC-entitled in work—which shifts the policy question towards the gates the model cannot exercise: the capital test and take-up among long-standing eligible non-claimants. Automatic enrolment after redundancy, contribution-based unemployment insurance and wage insurance are therefore candidate counterfactuals for future analysis, not policy recommendations established by the present results. Their costs and incidence require dedicated simulations and financing assumptions.

The repository also supplies reusable interfaces for the next policy exercises. Observable household and worker characteristics can tilt shock probabilities while preserving each sector's aggregate earnings loss, and a separate price-channel interface converts expenditure-weighted price changes into real disposable income. Neither interface supplies causal coefficients: the former requires externally estimated heterogeneity and the latter requires household expenditure data. They are extensions to populate and validate, not additional headline estimates in this paper.

6 Discussion and conclusion

The central finding is that the UK tax–benefit system responds differently when a common aggregate earnings loss is distributed through alternative labour-market margins. Across two loss anchors and three duration-equivalent scales, the wage-cut and displacement bounds differ by 7.3–8.0 percentage points; at the unit twelve-month scale the difference is 7.3 points, the wage-cut Exchequer cost is £83 million higher, and the household bootstrap places the difference at 7.6 points with a 95 per cent interval of 5.8–10.7 points in the record-resampling interval. The mechanism is institutional: proportional cuts receive tax relief near marginal rates, while whole-job loss receives relief nearer average rates and relies more on UC—relief that is gated by means tests, capital rules and claiming behaviour. Two sensitivity results sharpen this reading. Under a pension-gross income concept the cushioning *levels* fall by roughly four percentage points on both margins but the contrast is essentially unchanged, so the gap is a statutory asymmetry, not an artefact of the HBAI treatment of pension saving. And the take-up grid shows the gap is not an artefact of the claiming convention either: varying the new-entitlement rate from 0.55 to 1.00, or restoring the stale pre-shock flag, leaves the unit-stress displacement cushioning unchanged to within 0.1 points, because most displaced households are already UC-entitled in work. The contrast is pre-specified and conditional: it prices a statutory asymmetry, not the tariffs' total causal incidence. The annualisation of unemployment spells is conservative for it: the parameter-based bounding in Appendix K shows that correct monthly UC and partial-year PAYE accounting would raise displacement cushioning at sub-annual durations and leave the 12-month contrast exact.

Balanced integration makes the common aggregate loss comparable assignment by assignment and closely reproduces the Bernoulli mean at unit scale. It does not overcome sparse information: the unit displacement loss has only 6.2 effective record contributions, and support is weaker at the smallest OBR-duration cells. Assignment SDs describe scenario-allocation sensitivity; sampling uncertainty is scored separately by the household bootstrap, which is a sensitivity interval rather than a design-based confidence interval. The sign also survives all 23 leave-one-division-out reruns, with a cushioning difference of 5.3–9.6 points. This establishes sector robustness within the imposed bridge, not causal robustness of that bridge.

Calibrated longitudinal-LFS allocation sensitivities leave the central extensive-margin conclusion broadly unchanged: cushioning spans 36.0–37.4 per cent across cell, earnings-band and regularised-QRF risk shapes, compared with 36.8 per cent under uniform assignment. This robustness is conditional and limited. The imputation models agree weakly person by person, and the available LFS manufacturing target has only 177 donors; stability of an aggregate fiscal ratio is not evidence that tariff-specific worker selection has been identified.

National poverty changes remain near zero because the exposed population is small and often above the poverty line. They are not precise population estimates and do not imply that affected households' losses are small. EPD, inactivity, reallocation and demographic results are secondary or supplementary rather than competing headline estimands.

The common-loss transition-equivalent path makes the interior of the surface concrete: some workers receive an annualized re-employment penalty, some re-enter services at lower earnings, and survivors absorb the residual loss. Under the illustrative 70/30 transition allocation, the simulated cushioning is 41.8 per cent and the Exchequer cost is £464 million. These are conditional results: the annualized lag is encoded in earnings and hours rather than a monthly unemployment-benefit spell; the transition share, three-month lag and 28.3 per cent services penalty are scenario inputs, and the latter is a cross-sectional FRS destination gap that also captures hours and part-time composition. The response surface, including 50/50 and 85/15 allocations and the 14 per cent penalty, is therefore more informative than any single preferred point.

The principal limitation is that the analysis begins after the production-side transmission mechanism. It does not estimate how tariffs affect prices, quantities, output, value added, imported inputs, productivity, investment, profits, or labour demand. Instead, it maps a declared sector earnings loss directly onto households and evaluates the statutory response. The resulting estimates therefore identify fiscal incidence conditional on the imposed loss and adjustment path. They do not identify the economy-wide effect of the trade policy. Worker selection is independent within broad SIC divisions; the longitudinal LFS models provide only weakly concordant predictive rankings; within-year benefit timing, New Style JSA, household labour supply, indirect taxes, exchange rates, retaliation, and general equilibrium remain outside the central specification. Constituency results in Appendix A are illustrative because local industry structure is not observed at the required level.

The framework nevertheless provides a useful platform for subsequent work. The next empirical step is to combine product- or firm-level evidence on tariff exposure, prices, quantities, output, and worker outcomes with the household microsimulation. Such evidence would replace the imposed earnings bridge with an estimated transmission mechanism while retaining the adjustment margin and fiscal-incidence decomposition developed here.

The main conclusion is that, conditional on a common earnings loss, the labour-market margin materially changes the distribution of tax relief, benefit support, poverty effects, and public-finance costs. Wage-cut and displacement specifications are transparent bounds; the mixed and transition-equivalent paths show how intermediate adjustment patterns alter those outcomes. The results therefore speak directly to the fiscal consequences of labour-market adjustment, while leaving the size and composition of the underlying trade shock as an empirical question for future research.

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A Supporting Appendix

US exports use HMRC uktradeinfo OTS/RTS data for 2024, mapped from SITC to SIC 2007 divisions and divided by the latest non-suppressed ONS Annual Business Survey turnover. Of a £55,600 million customs-basis total, £52,500 million maps to divisions after documented exclusions, chiefly non-monetary gold and precious metals. The measured window uses May 2025–February 2026 against the mean of the same calendar months one and two years earlier. Its 12.7 per cent aggregate fall is descriptive, not causal.

B Shock mechanics

Each employee in exposed division j is independently selected with probability s_j. Weighted displaced counts and earnings therefore meet targets in expectation rather than exactly. Selected workers have employment income, hours, pension contributions and statutory pay zeroed. Wage cuts remove s_j of each exposed worker's earnings deterministically. Reallocation uses the same selected workers, assigns a services destination, and applies the calibrated earnings penalty.

UC claiming is redrawn after every margin for benefit units whose circumstances changed and that are entitled post-shock. The LCWRA override sets the benefit-unit element to the maximum of its stored baseline value and one statutory annual element, preventing duplicate payment.

C Monte Carlo and measures

Central direct and measured scenarios use 100 assignments. Every displayed $\pm$ value is an assignment SD, not a standard error or confidence interval. The machine-readable artifacts additionally store numerical Monte Carlo standard error as assignment SD divided by $\sqrt{n}$; it measures precision of the simulated mean only. Survey-design, tariff-response and model-parameter uncertainty are not estimated. Five-assignment mechanism, demographic, poverty-gap, reallocation, duration and sensitivity-grid exercises, and the ten-assignment supply-chain exercise, are exploratory and are excluded from the headline evidence; the take-up grid is scored at the full comparator draw count and reported in the main text. The mixed-adjustment scenario surface likewise uses five common assignments per cell. Its machine-readable cells and draws are in `results/scenario_testing.json` and `results/scenario_testing.csv`.

Reported relative BHC poverty changes hold the poverty line frozen at 60 per cent of the baseline equivalised median; absolute BHC poverty likewise uses 60 per cent of the baseline equivalised median as a fixed line. The cushioning rate is $1 - \Delta Y/\Delta E$, where ΔE is gross employment-earnings loss and ΔY is household disposable-income loss.

Thin-support cell. The OBR three-month duration-equivalent cell is omitted from the main bounds table because its displacement draw rests on 1.2 effective record contributions, with the largest record supplying 91.9 per cent of the loss on average. For completeness:

Anchor	Months	Margin	Gross loss (£m)	Exchequer (£m)	Cushioning (%)	Poverty (pp)
OBR	3	Displacement	89 ± 2	35 ± 8	30.9 ± 9.0	0.003
OBR	3	Wage cut	89	48	43.8	0.000

These rows are sensitivity evidence only, not population estimates.

Reconciling cushioning and the Exchequer effect. The Exchequer effect is the change in the aggregate simulated government balance, not the household-side transfer implied by the cushioning identity. The identity implies a household-side offset of $(1 - \Delta Y/\Delta E) \times \Delta E = \Delta E - \Delta Y$: for the full-schedule wage cut, $0.439 \times \pounds 886\text{m} \approx \pounds 389$ million, and for displacement $0.368 \times \pounds 882\text{m} \approx \pounds 325$ million, against reported Exchequer costs of £484 million and £412 million. The wedge is revenue lost outside employee disposable income: employer National Insurance contributions fall with the employee wage bill under every margin, and for displaced workers the zeroing of employee pension contributions and salary-sacrifice arrangements changes taxable pay relative to the gross-earnings accounting in ΔE. The government balance also absorbs any other simulated fiscal flows that respond to the shock. Cushion × gross therefore understates the Exchequer cost by roughly £87–95 million in the central full-schedule scenarios, and the two statistics answer different questions: the cushioning rate measures household income protection per pound of employee earnings lost, while the Exchequer effect measures the total first-round cost to the public finances.

D Elasticity and pass-through grid

The regenerated grid spans $\varepsilon \in \{0.4, 1, 2, 3\}$ and $\pi \in \{0.5, 0.75, 1\}$. Because only five displacement assignments are used per cell, it is an exploratory scaling diagnostic. Equal products $\varepsilon\pi$ generate equal deterministic wage-cut shocks; Bernoulli displacement outcomes vary across assignments. The grid is not an uncertainty interval.

E Factorial adjustment scenarios

Figure 1 separates the aggregate export-demand calibration from the assumed composition of labour-market adjustment. Each cell uses the full tariff schedule and five common assignments. The wage-cut endpoint is stable: cushioning is 40.6–41.3 per cent over the four calibrations. Moving towards displacement generally lowers cushioning, but the small-shock cells are especially noisy because a few high-weight FRS records dominate five Bernoulli assignments. These cells are stress scenarios, not estimates or confidence regions. The qualitative result is that fiscal cost scales mainly with the imposed gross shock, while the adjustment mix changes the share absorbed by taxes and transfers.

F Reallocation sensitivity

The reallocation exercise uses five paired assignments and is exploratory. Under the full schedule, the central 28.3 per cent penalty produces a mean gross loss of £264 million, Exchequer cost of £149 ± 131 million, cushioning of 39.7 ± 3.4 per cent, and zero measured BHC-poverty change. With a three-month lag, gross loss rises to £432 million and the Exchequer cost to £245 ± 216 million. The 14 per cent penalty gives gross loss of £131 million and Exchequer cost of £75 ± 64 million. These pound levels are not commensurable with the economy-wide wage-cut scenario because only selected movers' penalty is removed.

G Observed-outturn stress scenario

The descriptive May 2025–February 2026 export window falls 12.7 per cent against its two-year same-month baseline. This is not a causal tariff estimate: it includes prices, exchange rates, front-running, rerouting and other demand conditions.

The downside displacement scenario affects 17.3 thousand weighted workers on average, is cushioned by 36.6 ± 4.9 per cent and costs the Exchequer £419 ± 228 million. The wage-cut scenario is cushioned by 44.1 per cent and costs £517 million. This is an observed-outturn downside stress case, not a tariff-caused or preferred estimate; the validation artifact also reports signed net exposure before expanding divisions are clipped.

H HMRC destination-panel diagnostic

Across 1194 HS4 products, the capped value-weighted specification estimates a -33.9 per cent change in exports to the US relative to the same products sent to the three comparison destinations (95 per cent interval -47.8 to -16.4 per cent, $p =< 0.001$). Results have similar signs against each destination, and May 2019–23 pseudo-policy estimates are individually insignificant. This is not robust evidence of a common product effect: equal product weights give -3.2 per cent ($p = 0.519$), and restricting the sample to 2023 onward gives -21.1 per cent ($p = 0.121$). The estimated differential trend is itself borderline ($p = 0.046$). Most importantly, broad steel and auto chapters fall -22.5 per cent ($p = 0.283$), compared with -34.9 per cent for other products ($p = 0.001$): the tariff-intensity ordering fails. Figure 5 displays the monthly gap. The panel supports a post-policy decline concentrated in large US export products, but it cannot attribute that decline to tariffs or validate the tariff-rate mapping.

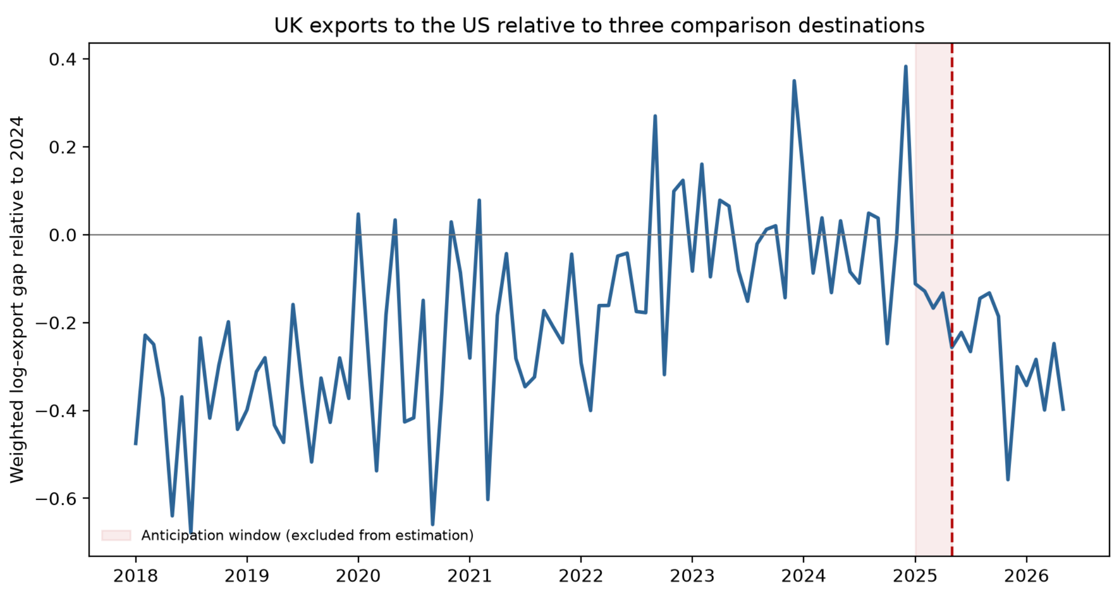

Figure 5: Monthly capped-value-weighted gap between UK HS4 exports to the United States and mean exports of the same products to Canada, Japan and Australia, normalised to January–March 2025.

The machine-readable specification, clustered standard errors, individual destination checks, sample-start sensitivities, tariff-intensity groups and May 2019–23 pseudo-policy estimates are stored in the event-study JSON artifact; the plotted observations are stored in its companion monthly CSV. Pre-policy value weights are capped at the 99th percentile, giving an effective product count of 107 despite 1194 included HS4 products. Uncapped and equal-weight estimates disclose the remaining concentration sensitivity.

I Demographic diagnostic

The gender, age and household-type breakdown uses five Bernoulli assignments and is exploratory. Selected exposed-sector workers are disproportionately men and concentrated in prime and older working ages, but individual subgroup estimates are too assignment-sensitive for precise rankings. The demographic artifact is therefore a diagnostic rather than an inferential table.

J Duration scope

The submission design reports three-, six- and twelve-month *duration-equivalent annual stresses*. It scales the probability of a coherent full-year displaced state rather than combining partial-year earnings with full-year unemployment status. These rows preserve expected worker-month loss but are not unemployment-spell simulations. Actual within-year tax, UC, waiting-period and re-employment timing requires a monthly model and is not claimed here.

K Monthly versus annual UC accounting

The annual model represents an m-month unemployment spell as a probability $m/12$ of a coherent full-year displaced state. This appendix bounds the resulting bias with statutory 2026 parameters for a representative single displaced worker aged 25 or over, without children, housing element or capital above the lower limit, at the exposed-sector FRS mean wage of £48,272. UC entitlement per worker-month is identical by construction: m months at the full standard allowance (£424.90 a month; earnings above the taper-out point in the working months) equals probability $m/12$ of twelve months. The differences run through income tax and timing. Because the tax schedule is convex, zeroing a full year of earnings forgoes relief at the worker's top marginal slices, whereas a partial-year earner sheds basic- and higher-rate slices: at this wage, the correct partial-year PAYE relief for a three-month spell is £2,414 against £1,785 in the annual equivalent, so the annual model *understates* total cushioning for a three- or six-month spell by about 5.2 percentage points (36.5 versus 31.3 per cent) and is exact at twelve months. The five-week UC payment wait shifts timing, not annual entitlement. The primary 12-month contrast in Table 1 is therefore unaffected, and the shorter duration-equivalent rows are conservative for the displacement margin: correcting the annualisation would raise displacement cushioning at those scales and narrow, not widen, the reported gap. Parameters and the full accounting are stored in `results/referee_fixes.json`.

L Supply-chain sensitivity

The input–output scenario adds upstream requirements using the ONS domestic Leontief inverse. The accounting shock has a 1.95 direct-plus-upstream factor. It is not a lower bound and is not multiplied by local-employment multipliers. In the exploratory ten-assignment microsimulation, displacement affects 29,700 weighted workers on average, costs the Exchequer £680 $\pm$ 310 million and raises relative BHC poverty by 0.028 percentage points. The wage-cut variant costs £960 million and leaves measured relative poverty unchanged. These figures are accounting sensitivities, not all-channel causal estimates.

M Constituency machinery

The enhanced-FRS constituency layer uses imputed industries rather than observed local industry structure. Its outputs are averaged over 20 paired worker-assignment draws (seeds 0–19), with each household's disposable-income change divided by household size before constituency means are constructed; the CSV artifact reports both the draw mean and the assignment-draw standard deviation. Because the constituency weights contain no observed local industry mix and the SIC imputation is held fixed, the outputs are synthetic demographic and income projections, not constituency exposure estimates. No map or seat-level figures are reported in this paper; the machinery is documented here so that future work can populate it with observed local industry data.